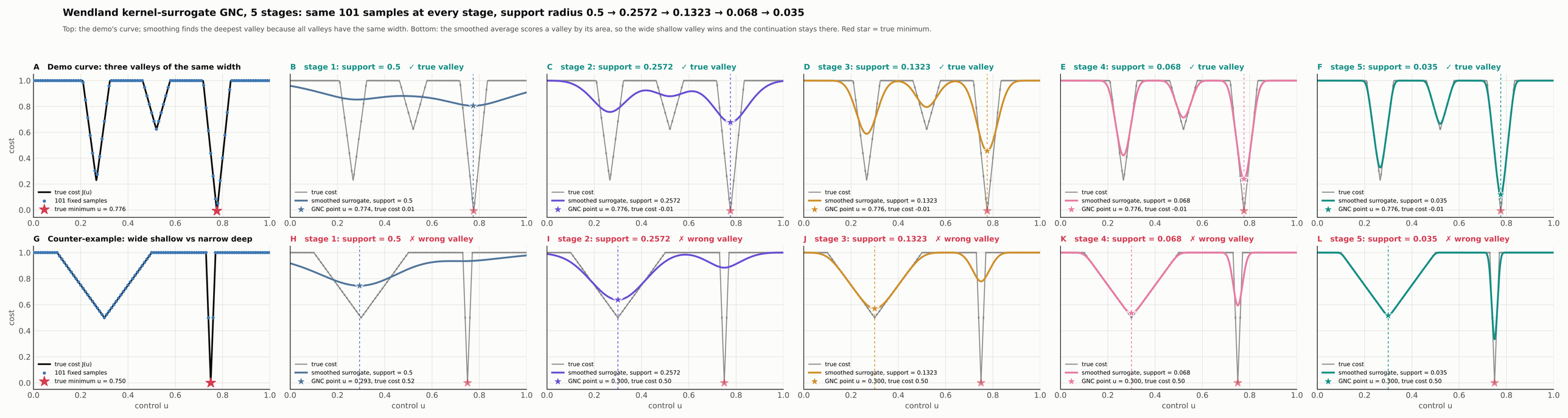

Wendland kernel-surrogate GNC, 5 stages: same 101 samples at every stage, support radius 0.5 → 0.2572 → 0.1323 → 0.068 → 0.035
Top: the demo's curve; smoothing finds the deepest valley because all valleys have the same width. Bottom: the smoothed average scores a valley by its area, so the wide shallow valley wins and the continuation stays there. Red star = true minimum.
A Demo curve: three valleys of the same width
B stage 1: support = 0.5 ✓ true valley
C stage 2: support = 0.2572 ✓ true valley
D stage 3: support = 0.1323 ✓ true valley
E stage 4: support = 0.068 ✓ true valley
F stage 5: support = 0.035 ✓ true valley
G Counter-example: wide shallow vs narrow deep
H stage 1: support = 0.5 ✗ wrong valley
I stage 2: support = 0.2572 ✗ wrong valley
J stage 3: support = 0.1323 ✗ wrong valley
K stage 4: support = 0.068 ✗ wrong valley
L stage 5: support = 0.035 ✗ wrong valley
cost
control u
true cost J(u)
101 fixed samples
true minimum u = 0.776
true minimum u = 0.750
true cost
smoothed surrogate, support = 0.5
GNC point u = 0.774, true cost 0.01
smoothed surrogate, support = 0.2572
GNC point u = 0.776, true cost -0.01
smoothed surrogate, support = 0.1323
GNC point u = 0.776, true cost -0.01
smoothed surrogate, support = 0.068
GNC point u = 0.776, true cost -0.01
smoothed surrogate, support = 0.035
GNC point u = 0.776, true cost -0.01
GNC point u = 0.293, true cost 0.52
GNC point u = 0.300, true cost 0.50